

Statement of Purpose

Machine Learning Summer School (MLSS) 2026

It all started with a simple ChatGPT prompt. I typed something random, got an answer back, and just sat there staring at the screen. Not because the answer was good but because it came back instantly. Something somewhere was processing language at a scale I could not even picture, and it did it in milliseconds. I needed to know how. That one question is what dragged me into machine learning, and I have not really come up for air since.

I started by doing what most people do. Following tutorials, running notebooks, copying architectures. But the moment things got real for me was when I tried building a brain tumor detection model from scratch. ResNet50, TensorFlow, Keras. I had the stack, I had the dataset, and I had a model that stubbornly sat at 71% validation accuracy and refused to move. I spent days on it. I found data leakage between my splits. I realized I had frozen the entire backbone and never thought to fine tune it. I had no proper evaluation metrics, no F1, no ROC AUC, nothing that would actually tell me if the model was useful in a clinical sense. Fixing all of that piece by piece was frustrating and slow and honestly the most I have ever learned from a single project. Around the same time I built PlansIQ, a mobile recharge comparison tool for Indian operators, packed into one HTML file with no backend and no external APIs. Just a working product I could hand someone and they would immediately get it.

Here is what I know about myself though. I am still missing the foundations. I can get a model running and I can debug it when it breaks. What I cannot do yet is explain with any real confidence why certain things work. Why does this optimizer converge faster here? What does this regularization actually guarantee? I have tried reading papers but I read them like a developer, not like a researcher. I skim for the method, miss the math, and move on. MLSS is where I want to fix that. Being around people who think rigorously about these things, researchers and peers who actually argue about theory, that is something I cannot replicate on my own.

The thing I would want anyone reading this to know is that I do not give up easily on things that confuse me. The tumor detection model could have been abandoned at 71%. It was not. I think that instinct, to stay with the hard thing until it makes sense, is what will make me genuinely useful in a research heavy environment like MLSS.

I am not applying because I am ready. I am applying because this is exactly the point where I need it.